Roll Number: _____	
Thapar Institute of Engineering and Technology, Patiala	
Department of Computer Science and Engineering	
BE (VIIth Semester) EST *19th July, 2025*	UCS802: Compiler Construction Time: 03 Hours; MM: 40 Name of Faculty: Dr. Geeta Kasana

Instruction: Attempt any four questions. out of five Attempt all parts of a question in a sequence. Draw neat diagram wherever required. *Assume any missing data logically.*

Q 1.	Consider the grammar given below: S→A|Y Y→ε A→ aA| b a) Generate the canonical LR(1) set of items for the grammar. b) Generate the ACTION and GOTO Table. c) Give the LALR set of items for the same	4+3+3
Q 2.	a) Rewrite the optimize code for the code given below after dead code elimination: **1. bar <- function(x) {** **2. y <- x * 2** **3. if (y < 0) {** **4. return("Negative")** **5. z <- 100** **6. }** **7. result <- y + 10** **8. temp <- 50** **9. return(result)** **10. temp <- temp + 10** **11. return(temp)** **12. }** **Note(1,2,3....are line numbers for reference you can use it to answer)** b) Differentiate between constant propagation and constant folding with the help of suitable examples. **c)** Draw a possible organization for the runtime environment (stack) if the user calls the function **fib(4)** to compute the 4th Fibonacci number using recursion. Also write down the code of the computed function.	2+3+5

Q3	Using suitable examples, differentiate between the following i) Synthesized and Inherited attributes ii) Dependency graph and Annotated Parse tree iii) LL and LR iv) L-attributed and S-attributed Grammar. v) SDT and SDD	2x5=10
Q 4.	a) Parse the following string "5 x 6 + 5 (8 + 6)" using Bottom-up evaluation of syntax directed translation and calculate the final value after evaluation of the string. b) Draw the DAG for the following expression: $x = (p + q) * (r - s) + (p + q) * t$ c) Determine the costs of the following instruction sequences with explanation: i)) *LD R0, Y* *LD R1, Z* *ADD R0, R0, R1* *ST x, R0* *LD R0, i* *MUL R0, R0,8* *LD R1, a(R0)* *ST b, R1*	4+4+2
Q 5.	a) Write the sequence of three-address instructions that the given expression would generate. $sum = sum + ((i \text{ and } j) \text{ or } (a \text{ and } b) \text{ and } (a \text{ or } b)$ b) Represent the Quadruple, Triple, and Indirect-Triple implementations for the above three-address code.	4+6

**********All the Best**********